

DATA-DRIVEN PROBABILITY MEASURES

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ABSTRACT. This paper presents a brief overview of data-driven probability measures, a surrogate AI model that estimates probabilities from data.

1. INTRODUCTION

In a standard binary classification task in machine learning, a deep neural network N is trained on features F and ground-truth labels $\{0, 1\}$, and the output $N(\bar{F})$ for an unseen feature $\bar{F}$ is often treated as a probability. However, because the output of $N(\bar{F})$ is a noisy prediction of 0 or 1, there is no rigorous mathematical or statistical guarantee that it constitutes a true probability. Here, we combine the Central Limit Theorem with standard binary classification to substitute a probability distribution with the distribution of machine learning outputs, constructing a data-driven probability surrogate AI model.

A key strength of the proposed model is that it can estimate the probabilities of the outcomes of the binary random variable

$$\Omega \ni \omega \mapsto X(\omega) \in \{0, 1\},$$

without assuming or explicitly specifying its underlying distribution since the Central Limit Theorem implies that the sample mean $\bar{X}^K$ converges in distribution to the standard normal distribution as K tends to infinity:

$$\bar{X}^K \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{K}}\right), \quad K \rightarrow \infty.$$

Thus, by estimating μ and σ using deep neural networks, we can calculate probabilities under normal distributions. In this paper, we present the construction of the proposed model and conduct several experiments. Note that the proposed model is a surrogate model and therefore provides an approximation of the “true” probability measure.

2. CONSTRUCTION

Let Ω be a sample space and X a random variable from Ω to $\{0, 1\}$. Define $\bar{X}^K$ as the sample mean of K draws of X :

$$\bar{X}^K = \frac{X_1 + X_2 + \cdots + X_K}{K}$$

By the Central Limit Theorem, when K is sufficiently large and each X_j is i.i.d., the distribution of $\bar{X}^K$ converges to a normal distribution. By the law of total probability:

$$\mathbb{P}(X = 1) = \sum_{i=0}^K \mathbb{P}\left(X = 1 \mid \bar{X}^K = \frac{i}{K}\right) \cdot \mathbb{P}\left(\bar{X}^K = \frac{i}{K}\right)$$

Since $X_i \in \{0, 1\}$ and i.i.d.:

$$\mathbb{P}\left(X = 1 \mid \bar{X}^K = \frac{i}{K}\right) = \frac{i}{K}$$

Therefore:

$$\mathbb{P}(X = 1) = \sum_{i=1}^K \frac{i}{K} \cdot \mathbb{P}\left(\bar{X}^K = \frac{i}{K}\right)$$

and

$$\bar{X}^K \sim \mathcal{N}(\mu, \sigma^2), \quad K \rightarrow \infty$$

To estimate $\mathcal{N}(\mu, \sigma^2)$, we substitute the distribution of $\bar{X}^K$ with that of deep learning outputs obtained via bootstrap sampling. Using bootstrap sampling from dataset $\mathcal{D}$, we derive multiple approximate functions $N^1, \dots, N^L$ via deep learning. For feature F , define the random variable $Z(F)$ as a discrete uniform distribution over $N^1(F), \dots, N^L(F)$:

$$Z(F) \sim \mathcal{U}(N^1(F), \dots, N^L(F))$$

The population mean and variance are:

$$\begin{aligned} \mu^{AI}(F) &= \frac{1}{L} \sum_{i=1}^L N^i(F), \\ \sigma^{AI}(F) &= \sqrt{\frac{1}{L} \sum_{i=1}^L (N^i(F) - \mu^{AI}(F))^2} \end{aligned}$$

The sample mean of $Z(F)$:

$$\bar{Z}^K(F) = \frac{Z_1(F) + Z_2(F) + \dots + Z_K(F)}{K}$$

converges by the Central Limit Theorem to:

$$\bar{Z}^K(F) \sim \mathcal{N}\left(\mu^{AI}(F), \frac{\sigma^{AI}(F)^2}{K}\right), \quad K \rightarrow \infty$$

Note that samples are not perfectly i.i.d., but bootstrap sampling practically reduces inter-model correlation. Since $\bar{X}^K$ is discrete while $\bar{Z}^K(F)$ is real-valued, we project $\bar{Z}^K(F)$ onto the grid $\{i/K\}_{i=1}^K$ to match the domain of $\bar{X}^K$, replacing $\mathbb{P}(\bar{X}^K = i/K)$ with the normalized mass:

$$\frac{\mathbb{P}\left(\bar{Z}^K(F) = \frac{i}{K}\right)}{\sum_{j=1}^K \mathbb{P}\left(\bar{Z}^K(F) = \frac{j}{K}\right)}$$

We therefore define the data-driven probability measure surrogate AI model as:

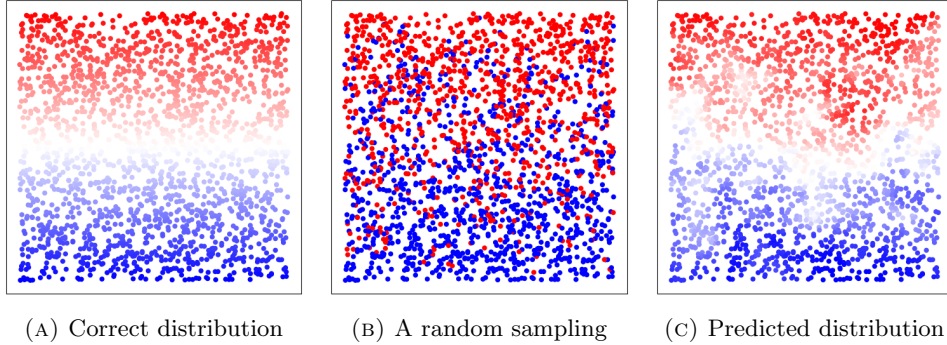
$$\mathbb{P}^{AI}(X = 1 \mid F, K) = \frac{\sum_{j=1}^K j \cdot \mathbb{P}\left(\bar{Z}^K(F) = \frac{j}{K}\right)}{K \cdot \sum_{j=1}^K \mathbb{P}\left(\bar{Z}^K(F) = \frac{j}{K}\right)}$$

3. PRELIMINARY EXPERIMENT 1

A scatter plot of points uniformly distributed over the unit square $[0,1] \times [0,1]$ in both x and y . Point color depends only on the y -coordinate, following a diverging red–white–blue colormap centered at $y = 0.5$: points near $y = 1$ are red, points near $y = 0$ are blue, and points near $y = 0.5$ are white/transparent. The x -coordinate has no effect on color. We use this to a verification of the proposal model.

This figure shows discrete samples (red/blue points) drawn directly from the probability distribution shown in the first image. At each point (x, y) , the color — red or blue — is sampled according to the probability defined by the original gradient: $P(\text{red})$ at that y -value equals the red intensity (alpha) in the first image, and $P(\text{blue}) = 1 - P(\text{red})$. As a result, points near the top are predominantly red, points near the middle are a mixture of red and blue, and points near the bottom are predominantly blue — reflecting the underlying probability at each y -value. This produces a noisy binary realization of the original continuous probability field, which can be used to train a model and then compare its recovered probability distribution against the true one used for sampling. In general, for the case of we do not know the original probability distribution (the first image), we would like to obtain the unknown probability distribution from a sample (the second image).

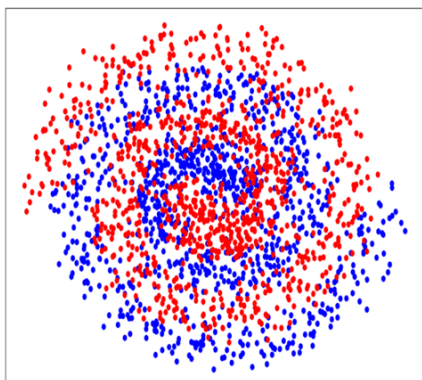
Using the random sampling data, we train the proposed model, and then we have the following as the probability distribution of the proposed model.



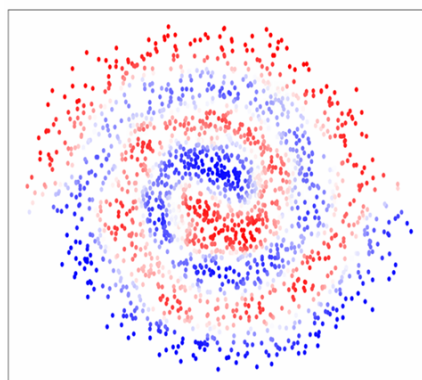
4. PRELIMINARY EXPERIMENT 2

This scatter plot shows a synthetic two-class dataset (red and blue points) generated for validating the probability estimation model. The points form a roughly circular cluster, with the two classes heavily overlapping near the center and becoming somewhat more separable toward the edges — providing a useful testbed where the true class probability varies smoothly rather than being strictly 0 or 1.

The prediction plot reveals a clear spiral structure that was not visually apparent in the original noisy scatter plot, with saturated red and blue arms winding from the center outward. Pale (low-confidence) points concentrate along the boundaries between arms — exactly where the true label should be most ambiguous — suggesting the model’s confidence estimate (σ^{AI}) is behaving sensibly. The center shows tighter, less resolved transitions, likely due to lower data density where the spiral winds most tightly.



(A) Original Data



(B) Prediction